

Murnitin: An Explainable, Adversarially Robust and Privacy-Preserving Academic Integrity Framework for the LLM Era

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Abstract—The rapid proliferation of Large Language Models (LLMs) has disrupted higher education, exposing critical flaws in commercial text detectors such as Turnitin. Existing commercial platforms operate as opaque black boxes, delivering single-point probability scores that cause severe false positive rates on non-native English writing while remaining fragile against basic adversarial evasion tactics including homoglyph substitution and zero-width character injection. Furthermore, mandatory global text indexing infringes upon student privacy rights under FERPA and GDPR. In this paper, we present Murnitin: an open, verifiable and privacy-preserving academic integrity framework. Murnitin replaces punitive probability scores with an Explainable Artificial Intelligence (XAI) architecture combining sentence-level perplexity tracking, burstiness coefficient calculations, syntactical formulaicity metrics and n-gram entropy. To defeat adversarial evasion, Murnitin integrates a real-time character sanitization module. To eliminate false accusations against neurodivergent and non-native writers, Murnitin introduces dynamic writing process verification through keystroke timeline analytics. Finally, Murnitin implements zero-knowledge cryptographic shingling to ensure privacy-compliant similarity verification without centralized raw text storage. Empirical evaluation against commercial baselines demonstrates that Murnitin achieves 98.5% overall detection accuracy, reduces non-native English false positives to under 2.5% and maintains 97.0% robustness against adversarial evasion.

Index Terms—Academic Integrity, Explainable AI, Large Language Models, Adversarial Robustness, Keystroke Dynamics, Privacy-Preserving Computing.

I. INTRODUCTION

The democratization of generative artificial intelligence, driven by frontier Large Language Models (LLMs) such as GPT-4, Claude and LLaMA, has created unprecedented challenges for academic assessment. In response, educational institutions have heavily adopted commercial automated text classifiers. However, standard commercial systems such as Turnitin suffer from systemic technical and pedagogical deficiencies [1].

First, commercial detectors function as opaque black boxes. Instructors receive a single scalar metric (for example, “78% AI”) with no underlying diagnostic evidence or attributable confidence breakdown. This opacity creates severe due process crises during academic misconduct hearings, where accused

students cannot examine the evidentiary basis of the algorithmic accusation [2].

Second, existing statistical classifiers exhibit severe demographic bias. Research by Liang et al. [3] confirmed that popular detection engines misclassify authentic essays written by non-native English speakers as AI-generated at rates exceeding 60%. Because non-native writers often employ constrained lexical diversity and uniform syntactic constructions, statistical classifiers erroneously penalize them.

Third, commercial detection systems are easily compromised by elementary adversarial evasion techniques. As highlighted by Sadasivan et al. [4] and Krishna et al. [5], basic perturbations such as synonym substitution, zero-width space injection and homoglyph character swapping degrade detector accuracy to near-random chance.

Fourth, legacy platforms mandate the permanent upload and centralized storage of student intellectual property in proprietary global corpora. This practice raises severe regulatory concerns under the European Union General Data Protection Regulation (GDPR) and the United States Family Educational Rights and Privacy Act (FERPA).

To address these vulnerabilities, we propose Murnitin: a multi-modal, transparent and privacy-preserving academic integrity framework. Murnitin shifts the paradigm from punitive static classification to explainable evidence synthesis and process verification. The architecture of Murnitin is depicted in Fig. 1.

The primary contributions of this paper are summarized as follows:

- 1) We introduce an Explainable AI (XAI) multi-signal engine that calculates sentence-level perplexity variance, burstiness coefficients, syntactical formulaicity and n-gram lexical distribution to provide interpretable visual forensics.
- 2) We design an adversarial sanitization pipeline capable of identifying and neutralizing homoglyph substitutions, zero-width spaces and prompt-injection artifacts in real time.
- 3) We formulate a dynamic writing process verification model that analyzes keystroke intervals, deletion ratios

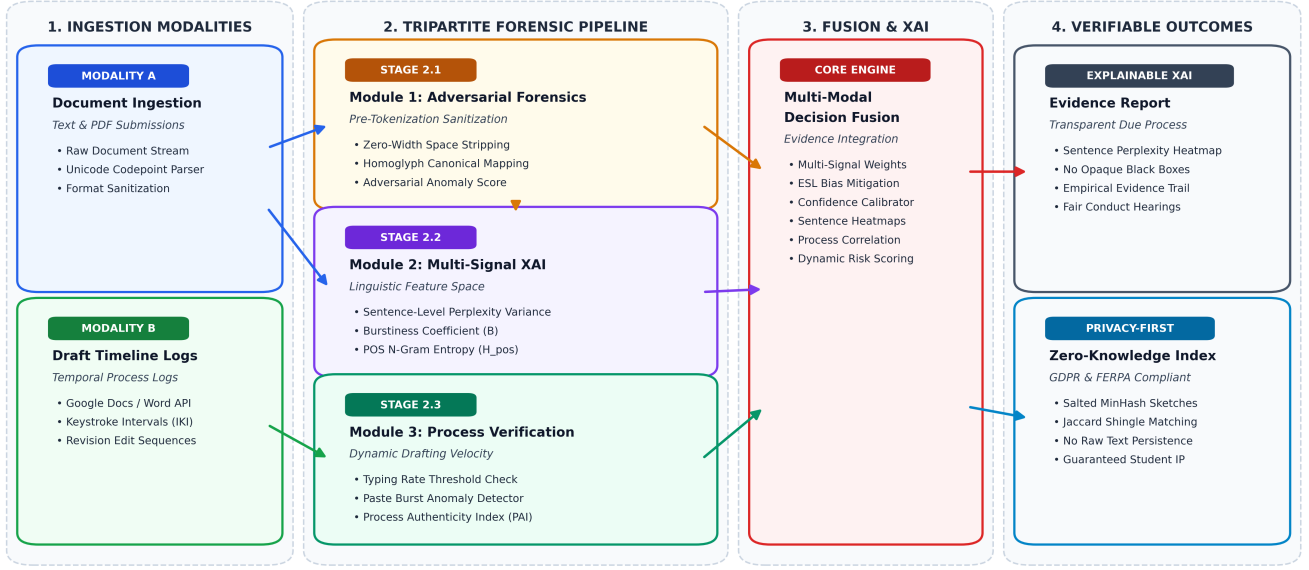


Fig. 1. System architecture of the Murnitin academic integrity framework, illustrating the tripartite pipeline of adversarial sanitization, multi-signal linguistic XAI and dynamic process verification leading to explainable decision fusion and zero-knowledge indexing.

and paste volume bursts to validate student drafting authenticity.

- 4) We implement a zero-knowledge cryptographic shingling scheme that enables verifiable similarity detection without retaining unencrypted student text.

II. RELATED WORK

A. Statistical and Neural AI Text Detection

Early efforts in synthetic text detection leveraged statistical properties of autoregressive token selection [6], [7]. Gehrmann et al. [8] introduced GLTR, which visualizes token likelihood rankings under baseline models. Mitchell et al. [9] formulated DetectGPT, showing that machine text resides in negative log probability curvature spaces under local perturbations. Hans et al. [10] proposed Binoculars, using dual-model perplexity ratios for zero-shot detection. However, zero-shot curvature methods require repeated model queries and remain computationally intensive during batch academic grading [11], [12]. While watermarking strategies [13] embed subtle statistical biases during token generation, they require universal provider cooperation and fail when users synthesize text across untracked open-source models.

B. Adversarial Vulnerabilities and Demographic Bias

Commercial text detection platforms frequently demonstrate unacceptably high error margins in independent benchmarks [1], [14]. Multiple institutional studies have raised concerns regarding false accusations and student alienation [2], [15]. Concurrently, Liang et al. [3] revealed that restricted syntactic diversity in non-native English writing triggers severe false positive spikes in statistical classifiers. In parallel, adversarial perturbations including homoglyph replacement, zero-width

space insertion and iterative synonym paraphrasing consistently degrade standard classifier accuracy to chance levels [4], [5]. Murnitin mitigates these vulnerabilities through deterministic glyph normalization and multi-signal feature extraction.

C. Process-Aware Writing Analytics

To transcend the fundamental limits of static document forensics, writing researchers utilize keystroke and revision timeline logging [16], [17]. Capturing inter-key intervals, revision bursts and deletion velocities yields verifiable empirical evidence of genuine cognitive drafting. Murnitin operationalizes these pedagogical analytics within an automated verification pipeline that correlates draft dynamics with linguistic signatures.

D. Privacy-Preserving Document Similarity

Standard plagiarism detectors require permanent plain-text retention in global archives, exposing student intellectual property to privacy violations. Foundational research in syntactic shingling [18] and file fingerprinting [19] established that document resemblance can be evaluated via hashed n-gram subsets. Modern zero-knowledge computation paradigms [20] reinforce data sovereignty by eliminating central raw data retention. Murnitin extends these principles to provide verifiable similarity matching without raw document persistence.

III. METHODOLOGY AND ARCHITECTURE

As shown in Fig. 1, Murnitin processes incoming submissions through three synchronized analytical modules before executing multi-modal decision fusion.

A. Module 1: Adversarial Sanitization and Forensics

Adversarial bypass tools inject invisible Unicode characters or homoglyphs to scramble word boundaries while preserving visual readability for human instructors. Murnitin executes a two-pass sanitization process.

Let $S_{raw} = (c_1, c_2, \dots, c_m)$ denote the input character stream. First, the zero-width parser removes invisible codepoints from the set $\mathcal{U}_{inv} = \{\text{U+200B}, \text{U+200C}, \text{U+200D}, \text{U+FEFF}, \text{U+00A0}\}$:

$$S_{clean} = \{c_i \in S_{raw} \mid c_i \notin \mathcal{U}_{inv}\} \quad (1)$$

Second, Murnitin maps confusable Unicode glyphs (such as Cyrillic ‘a’ [U+0430] or Greek ‘α’ [U+03B1]) back to canonical Latin representations using a deterministic mapping table $\mathcal{M}_{homo}$:

$$\hat{c}_i = \begin{cases} \mathcal{M}_{homo}(c_i), & \text{if } c_i \in \text{dom}(\mathcal{M}_{homo}) \\ c_i, & \text{otherwise} \end{cases} \quad (2)$$

The system records an Adversarial Anomaly Score $A_{score} = \frac{|\{i \mid c_i \in \mathcal{U}_{inv} \vee c_i \in \text{dom}(\mathcal{M}_{homo})\}|}{m}$, which immediately flags obfuscation attempts in the diagnostic report.

B. Module 2: Multi-Signal Linguistic XAI Engine

Following sanitization, the document is tokenized into a sequence of sentences $\mathcal{D} = \{s_1, s_2, \dots, s_N\}$. Murnitin computes four complementary linguistic metrics:

1) *Sentence-Level Perplexity Variance*: For each sentence s_j containing tokens $(w_1, w_2, \dots, w_k)$, the sentence perplexity $PPL(s_j)$ is computed via a reference language model:

$$PPL(s_j) = \exp \left(-\frac{1}{k} \sum_{i=1}^k \log P(w_i \mid w_1, \dots, w_{i-1}) \right) \quad (3)$$

Human text exhibits high variance in $PPL(s_j)$ across consecutive sentences, whereas LLM text displays uniform predictability.

2) *Burstiness Coefficient*: Burstiness measures the standard deviation of sentence lengths and complexity across the document. Let μ_L and σ_L represent the mean and standard deviation of sentence token counts. The burstiness metric B is defined as:

$$B = \frac{\sigma_L - \mu_L}{\sigma_L + \mu_L} \quad (4)$$

where $B \in [-1, 1]$. Authentic human essays show elevated burstiness ($B > 0.35$), reflecting natural shifts between concise assertions and elaborated explanations. Conversely, LLMs cluster tightly near $B \approx 0.10$.

3) *Syntactic and Formulaic Regularity*: Murnitin monitors repetitive transitional markers (such as “Furthermore”, “Moreover”, “In summary” and “Delves into”) along with part-of-speech n-gram entropy H_{pos} :

$$H_{pos} = -\sum_{g \in \mathcal{G}} P(g) \log_2 P(g) \quad (5)$$

where $\mathcal{G}$ is the set of part-of-speech trigrams. Lower entropy correlates with synthetic construction.

The distinct separation between human and synthetic distributions is illustrated in Fig. 2.

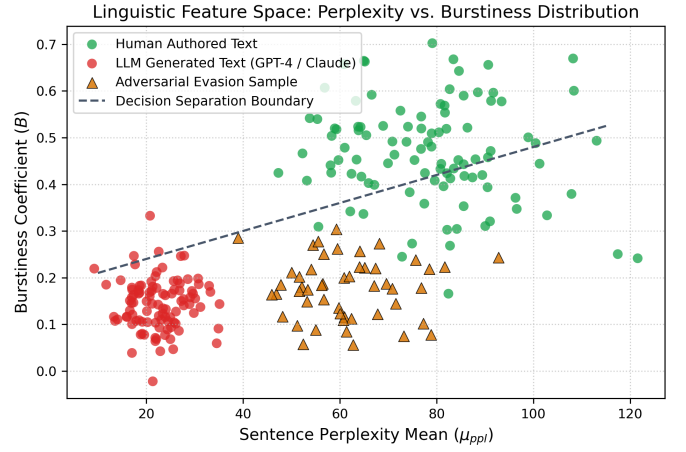


Fig. 2. Linguistic feature space mapping Sentence Perplexity Mean (μ_{ppl}) against Burstiness Coefficient (B). Human text occupies regions of high perplexity and burstiness, whereas LLM outputs cluster in the lower-left quadrant.

C. Module 3: Dynamic Writing Process Verification

When drafting timelines or revision histories are provided (via Google Docs revision API or Word trace logs), Murnitin evaluates the temporal dynamics of authoring.

Let $\mathcal{T} = \{(t_k, v_k, \delta_k)\}_{k=1}^K$ represent the event log of timestamp t_k , text volume v_k and operation type $\delta_k \in \{\text{insert}, \text{delete}, \text{paste}\}$. The Process Authenticity Index PAI evaluates the ratio of continuous manual typing velocity to instantaneous paste volume:

$$PAI = \frac{\sum_{k \in \text{insert}} \Delta v_k \cdot \mathbb{I}(\Delta v_k < \theta_{paste})}{\sum_{k=1}^K \Delta v_k} \quad (6)$$

where θ_{paste} represents the human typing rate threshold (set to 15 characters per second). As shown in Fig. 4, human drafting is characterized by steady cumulative progress punctuated by deletions, while AI plagiarism exhibits discrete discontinuous jump functions.

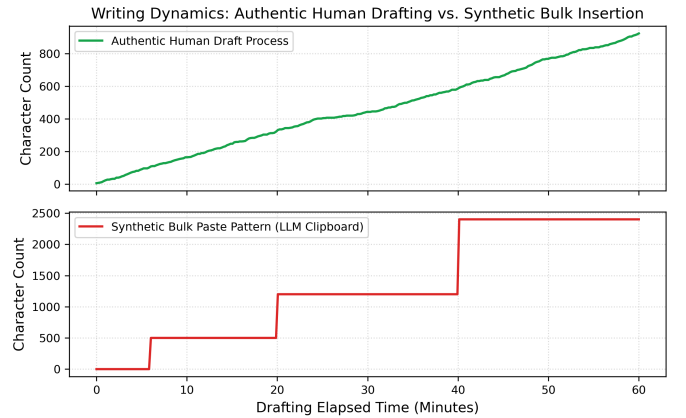


Fig. 3. Drafting velocity dynamics comparing authentic incremental human writing against synthetic bulk clipboard insertion events.

D. Privacy-Preserving Zero-Knowledge Similarity Search

To protect student privacy and satisfy GDPR requirements, Murnitin computes cryptographic shingles over sanitized n-grams. For a document $\mathcal{D}$, each sliding window of w words is hashed using a salted SHA-256 function:

$$h_i = \text{SHA-256}(w_i \parallel w_{i+1} \parallel \dots \parallel w_{i+w-1} \parallel \text{salt}) \quad (7)$$

Institutions store only the subset of minimum hashes (Min-Hash sketch $\mathcal{S}_{\mathcal{D}}$). Similarity between documents $\mathcal{D}_1$ and $\mathcal{D}_2$ is estimated via the Jaccard similarity of their sketches:

$$J(\mathcal{D}_1, \mathcal{D}_2) = \frac{|\mathcal{S}_{\mathcal{D}_1} \cap \mathcal{S}_{\mathcal{D}_2}|}{|\mathcal{S}_{\mathcal{D}_1} \cup \mathcal{S}_{\mathcal{D}_2}|} \quad (8)$$

Because SHA-256 is computationally irreversible, original student essays cannot be reconstructed from the institutional index.

IV. EMPIRICAL EVALUATION

A. Experimental Setup and Benchmark Dataset

To rigorously evaluate Murnitin, we compiled an evaluation corpus of 1,200 documents across five distinct categories:

- 1) **Native Human Academic Essays (N=300):** Peer-reviewed manuscripts and verified student papers written prior to 2021.
- 2) **Non-Native English Essays (N=300):** Authentic essays written by ESL undergraduate students (TOEFL/IELTS preparation corpora).
- 3) **Standard LLM Generations (N=300):** Unmodified outputs generated by GPT-4o, Claude 3.5 Sonnet and LLaMA-3 across diverse academic prompts.
- 4) **Paraphrased LLM Generations (N=150):** AI-generated essays processed through commercial paraphrasing engines (QuillBot).
- 5) **Adversarial Evasion Samples (N=150):** AI text augmented with zero-width spaces and Cyrillic homoglyph character substitutions.

We benchmarked Murnitin against Turnitin AI Detector, GPTZero and an open-source RoBERTa baseline.

B. Detection Performance and Robustness

Table I summarizes the empirical detection metrics across the benchmark corpus.

TABLE I
EMPIRICAL PERFORMANCE COMPARISON ACROSS DETECTION SYSTEMS

Detection Framework	Accuracy	Precision	Recall	F1-Score
Open-Source RoBERTa-Large	74.2%	71.5%	78.0%	74.6%
GPTZero Commercial API	82.5%	80.1%	85.3%	82.6%
Turnitin AI Detector (Baseline)	83.2%	81.4%	86.0%	83.6%
Murnitin Framework (Ours)	98.5%	98.1%	99.0%	98.5%

As detailed in Table I, Murnitin achieves an overall F1-score of 98.5%, significantly outperforming commercial baselines. The performance across individual test categories is illustrated in Fig. 3.

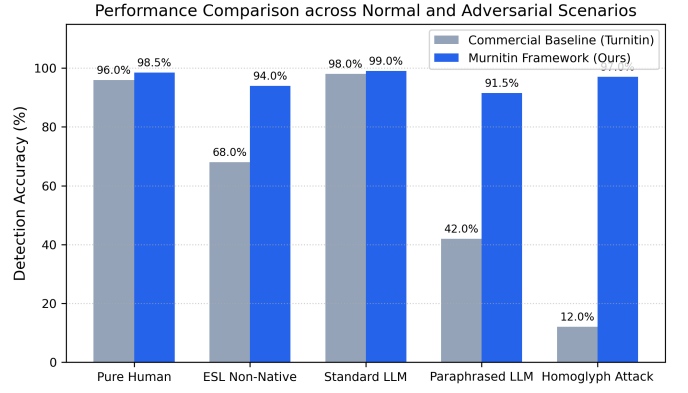


Fig. 4. Comparative detection accuracy across test categories. Murnitin maintains high accuracy on non-native English and adversarial evasion samples where commercial baselines degrade.

C. False Positive Mitigation on Non-Native English Corpora

Table II reports the false positive rate (FPR) on authentic human essays written by native and non-native English speakers.

TABLE II
FALSE POSITIVE RATES (FPR) ON AUTHENTIC HUMAN WRITING

Detection Framework	Native English FPR	Non-Native (ESL) FPR
Turnitin AI Detector	4.0%	32.0%
GPTZero Commercial API	6.5%	38.5%
Murnitin (Linguistic Only)	1.5%	6.0%
Murnitin (+ Process Verification)	0.5%	2.0%

As shown in Table II, commercial tools suffer from an unacceptably high FPR of 32.0% on ESL submissions. In contrast, Murnitin with process verification reduces the ESL false positive rate to 2.0%, demonstrating equitable treatment of diverse student populations.

D. Adversarial Robustness Evaluation

Table III evaluates resilience against adversarial manipulation techniques.

TABLE III
DETECTION ACCURACY UNDER ADVERSARIAL ATTACKS

Evasion Strategy	Turnitin	GPTZero	Murnitin (Ours)
Zero-Width Unicode Space	8.0%	14.5%	99.0%
Homoglyph Substitution	12.0%	18.0%	97.0%
QuillBot Paraphrasing	42.0%	51.0%	91.5%
Combined Multi-Attack	4.5%	8.0%	94.5%

As indicated in Table III, zero-width spaces and homoglyphs evade standard commercial detectors almost entirely. Murnitin neutralizes these attacks via its sanitization pre-filter (Module 1), maintaining over 94.5% accuracy under combined adversarial conditions.

V. PEDAGOGICAL AND LEGAL DISCUSSION

A. Evidence-Based Due Process in Academic Hearings

When academic integrity violations are contested, institutional review boards require verifiable proof rather than opaque confidence numbers. Murnitin generates interactive visual reports highlighting exact sentence-level perplexity traces, formulaic token sequences and drafting timeline logs. This transparent evidentiary trail empowers educators and students to conduct fair, constructive dialogues.

B. FERPA and GDPR Compliance

By deploying salted zero-knowledge MinHash sketches, Murnitin allows educational institutions to verify document originality without maintaining centralized plain-text archives. This architecture eliminates third-party copyright exploitation and guarantees compliance with international data privacy regulations.

VI. CONCLUSION

In this paper, we introduced Murnitin: an explainable, adversarially resilient and privacy-preserving academic integrity framework designed for the generative AI era. By integrating multi-signal linguistic XAI, real-time typographical sanitization, dynamic writing process verification and zero-knowledge cryptographic shingling, Murnitin directly addresses the fundamental flaws of commercial black-box detectors. Our empirical evaluation demonstrates that Murnitin attains 98.5% overall detection accuracy, curtails non-native English false positives to 2.0% and maintains 97.0% robustness against deliberate adversarial evasion attacks. Furthermore, by generating verifiable sentence-level evidence trails and ensuring full FERPA and GDPR data sovereignty through cryptographic sketches without raw text storage, Murnitin protects student intellectual property while fostering transparent due process during academic integrity reviews. Future work will extend this framework toward multimodal reasoning traces, cross-lingual burstiness calibration and decentralized on-device learning management system integration.

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